

Lab13 – Custom Resource Definition

1. Create a CRD resource named `backup.example.com` with the following specification :

- Group: `example.com`
- Version: `v1`
- Kind: `Backup`
- Singular: `backup`
- Plural: `backups`
- Properties of type `string`: `cronExpression`, `podName`, `path`

```
vagrant@kubernetes-control-plane:~/Lab$ cat backup-resource.yaml
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: backups.example.com
spec:
  group: example.com
  versions:
  - name: v1
    served: true
    storage: true
    schema:
      openAPIV3Schema:
        type: object
        properties:
          spec:
            type: object
            properties:
              cronExpression:
                type: string
              podName:
                type: string
              path:
                type: string
        scope: Namespaced
      names:
        kind: Backup
        singular: backup
        plural: backups
        shortNames:
        - bu
vagrant@kubernetes-control-plane:~/Lab$
```

```
vagrant@kubernetes-control-plane:~/Lab$ kubectl apply -f backup-resource.yaml
customresourcedefinition.apiextensions.k8s.io/backups.example.com created
vagrant@kubernetes-control-plane:~/Lab$
```

2. Retrieve the details for the `Backup` custom resource created in the previous step.

```
vagrant@kubernetes-control-plane:~/Lab$ kubectl get crd backups.example.com
NAME                CREATED AT
backups.example.com 2024-08-25T13:06:41Z
vagrant@kubernetes-control-plane:~/Lab$ kubectl describe crd backups.example.com
Name:                backups.example.com
Namespace:
Labels:              <none>
Annotations:         <none>
API Version:         apiextensions.k8s.io/v1
Kind:                CustomResourceDefinition
Metadata:
  Creation Timestamp: 2024-08-25T13:06:41Z
  Generation:        1
  Resource Version:   12089
  UID:               4a614884-4480-419b-9bad-468b057f23fa
Spec:
  Conversion:
    Strategy: None
  Group:      example.com
  Names:
    Kind:      Backup
    List Kind: BackupList
    Plural:    backups
    Short Names:
      bu
    Singular:  backup
```

3. Create a custom object named `nginx-backup` for the CRD. Provide the following property values:

- `cronExpression`: `0 0 \* \* \*`
- `podName`: `nginx`
- `path`: `/usr/local/nginx`

```
vagrant@kubernetes-control-plane:~/Lab$ cat backup.yaml
apiVersion: example.com/v1
kind: Backup
metadata:
  name: nginx-backup
spec:
  cronExpression: "0 0 * * *"
  podName: nginx
  path: /usr/local/nginx

vagrant@kubernetes-control-plane:~/Lab$ kubectl apply -f backup.yaml
backup.example.com/nginx-backup created
vagrant@kubernetes-control-plane:~/Lab$
```

4. Retrieve the details for the `nginx-backup` object created in the previous step.

```
vagrant@kube-control-plane:~/Lab$ kubectl get backup
NAME          AGE
nginx-backup  96s
vagrant@kube-control-plane:~/Lab$
vagrant@kube-control-plane:~/Lab$ kubectl get backups
NAME          AGE
nginx-backup  99s
vagrant@kube-control-plane:~/Lab$
vagrant@kube-control-plane:~/Lab$ kubectl get bu
NAME          AGE
nginx-backup  103s
vagrant@kube-control-plane:~/Lab$
vagrant@kube-control-plane:~/Lab$ kubectl describe bu nginx-backup
Name:          nginx-backup
Namespace:     default
Labels:        <none>
Annotations:   <none>
API Version:   example.com/v1
Kind:          Backup
Metadata:
  Creation Timestamp:  2024-08-25T13:13:51Z
  Generation:         1
  Resource Version:    12735
  UID:                 5c7f2cb8-b3d3-4d68-9139-9d37ba45f299
Spec:
  Cron Expression:    0 0 * * *
  Path:               /usr/local/nginx
  Pod Name:          nginx
Events:              <none>
vagrant@kube-control-plane:~/Lab$
```